

LITERATURE SURVEY

SL. NO	TITLE	YEAR OF PUBLISHING	TECHNOLOGY USED	CONCLUSION
1	An Enhanced YOLOv8 Model for Real-Time and Accurate Pothole Detection and Measurement.	2025	YOLOv8n-seg, Dynamic Snake Convolution (DSConv), Simple Attention Module (SimAM), GELU activation.	They created the PothRGBD RGB-D dataset and proposed a lightweight YOLOv8n-seg variant that improves segmentation and enables accurate perimeter & depth estimation. Therefore, I can conclude from their paper that integrating DSConv and SimAM into a YOLOv8 segmentation yields more accurate pothole masks and reliable physical measurements, which I can apply to my pothole-detection project to get precise segmentation masks and direct depth/perimeter estimates.
2	Robust Video-Based Pothole Detection and Area Estimation for Intelligent Vehicles with Depth Map and Kalman Smoothing	2025	ACSH-YOLOv8 BoT-SORT, Ego-motion compensation, DepthAnything, MBTP, CDKF (Confidence–Distance Kalman Filter).	They present an end-to-end vision pipeline to achieve accurate and stable pothole area estimation from video frames. I conclude from their paper that integrating enhanced object detection, monocular depth estimation, pixel-level area computation, and Kalman-based smoothing can significantly improve the precision and reliability of pothole detection.
3	Mobile3DRecon: Real-time Monocular 3D Reconstruction on a Mobile Phone	2020	Monocular camera, 6DoF SLAM system, Semi-global matching (SGM), DNN-based depth refinement, Scalable TSDF voxel fusion, Mesh generation algorithm on mobile CPUs.	They presented a mobile system capable of generating dense, real-time 3D surface meshes using only an embedded monocular camera. They show robust results for AR and object interaction on mid-range phones. Therefore, I can conclude that their mobile-focused, CPU-based incremental mesh generation offers techniques and system design ideas directly applicable to resource-constrained 3D reconstruction tasks like pothole mapping or detection.

4	Pothole and Plain Road Classification Using Adaptive Mutation Dipper Throated Optimization and Transfer Learning for Self-Driving Cars.	2022	ResNet-50, Binary Advanced Mutation, Dipper Throated Optimization, Optimized hashing SMOTE, Random Forest (RF), AMDTO optimizer.	This study showed a pipeline that combines transfer-learning feature extraction. Therefore, I can conclude from their paper that applying meta-heuristic optimization for feature selection and classifier tuning together with an improved SMOTE balancing step can dramatically boost pothole vs plain-road classification accuracy.
5	Real-time scene reconstruction and triangle mesh generation using multiple RGB-D cameras	2017	Point cloud smoothing, Mesh zippering, Direct triangle mesh generation, Mesh merging.	They proposed a real-time 3D reconstruction method with a GPU-optimized mesh merging method. They demonstrate both high mesh quality and significant acceleration compared to CPU-based methods. I can conclude that their GPU-parallel mesh approach, especially the MLS point cloud smoothing and mesh merging, gives insight into achieving fast, reliable surface modeling—knowledge that is valuable for enhancing the efficiency of mesh processing in pothole detection systems.